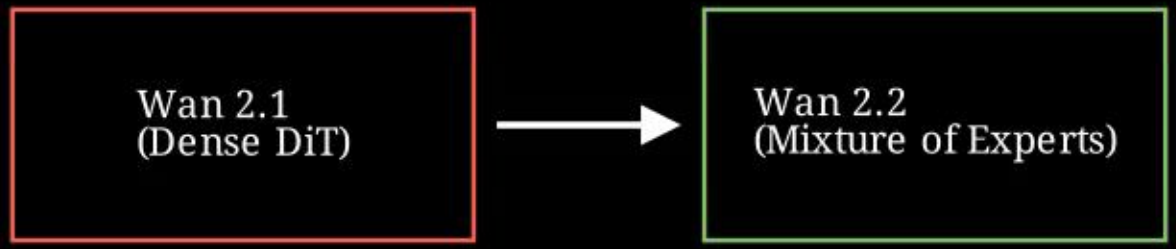
# Long-Form Reference-to-Video Generation

## Wan Animate Progress Report

By: Natnael.Y

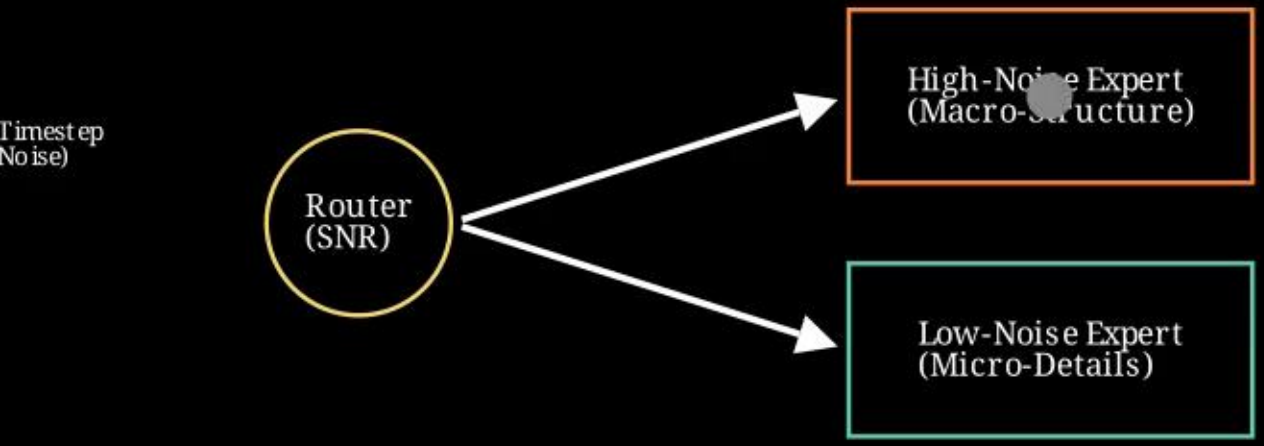
# Architecture Shift: Migrating to Wan 2.2




- All parameters compute every step
- Heavy compute bottleneck

- Sparse routing
- 27B Parameters (Only 14B Active)

# How MoE Routes Diffusion Steps



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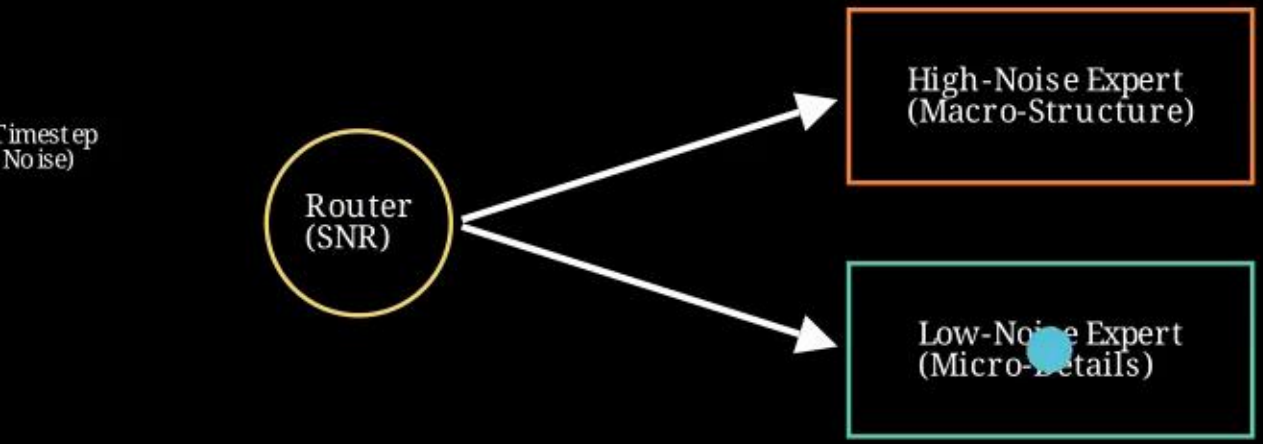

# Breaking the Length Barrier & Memory Management

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- Seamless Transitions: Wanlooper divides videos, blending via Hamming Windo
- Memory Constraints: Fixed 'Process Killed' OOM error by controlling RAM load

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Smooth Audio

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- Lip-Sync: Integrated MultiTalk model for coherent mouth synchronization.
- Audio Coherence: Fixed spikes/pops by deferring audio merging to final step

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- Foundation: Built upon Wan-I2V, enhanced with Mixture of Experts (MoE).

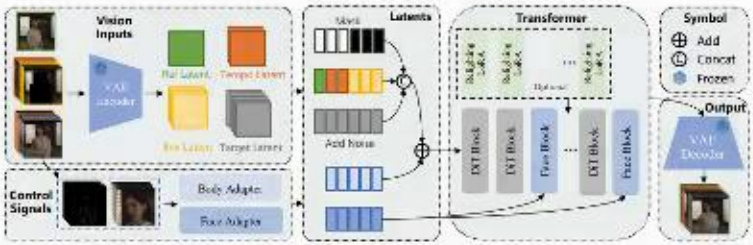


Figure 2: Overview of *Wan-Animate*, which is built upon Wan-I2V. We modify its input formulation to unify reference image input, temporal frame guidance, and environmental information (for dual-mode compatibility) under a common symbolic representation. For body motion control, we use skeleton signals that are merged via spatial alignment. For facial expression control, we leverage implicit features extracted from face images as the driving signal. Additionally, for character replacement, we train an auxiliary Relighting LoRA to enhance the character's integration with the new environment.

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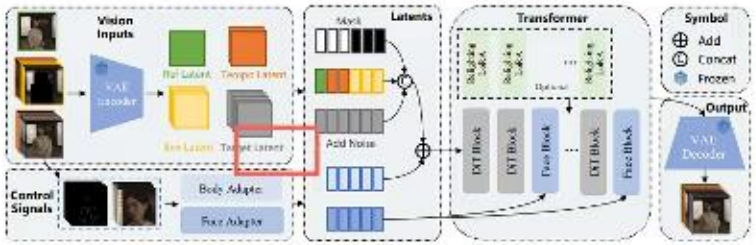


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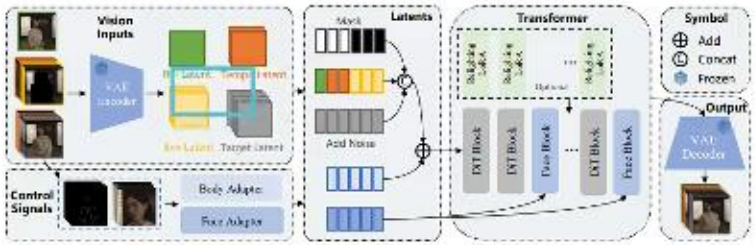


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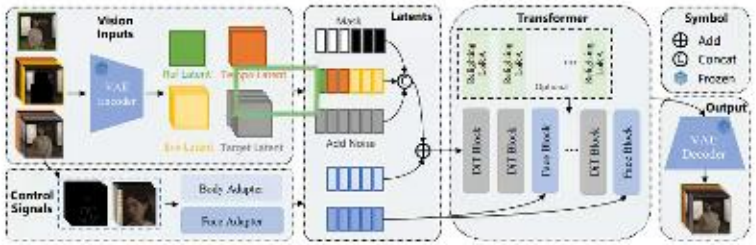


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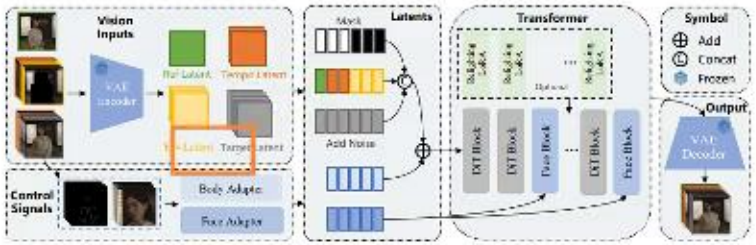


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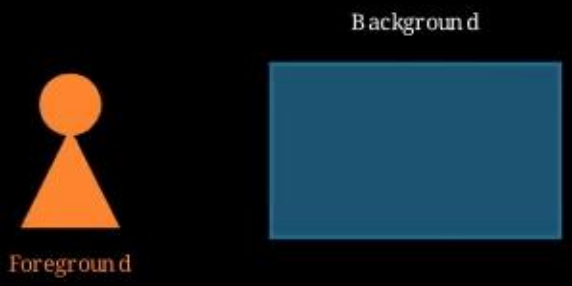

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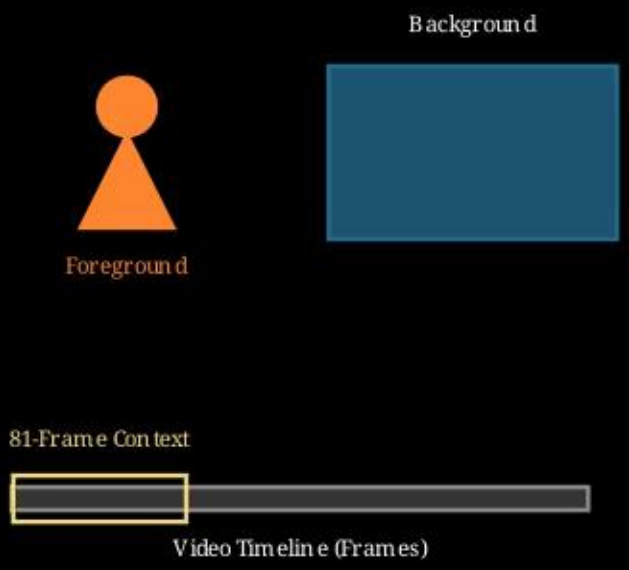
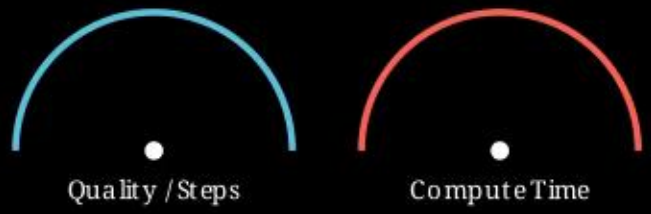
- Segment Interpolation: Incorporated the RIFE VFI Interpolation node to resolve frame-to-frame coherence.
- Background Extraction: Integrated Sam2Segmentation and custom masking to surgically extract backgrounds.
- Lighting Stability: Utilized a sliding window technique (81-frame context) to maintain constant lighting.



# Quality vs. Compute Optimization

- The Trade-Off: Increased steps (5 -> 21) boosted clarity but pushed time to 4h 20m.



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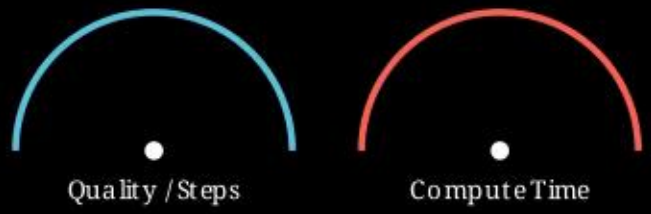

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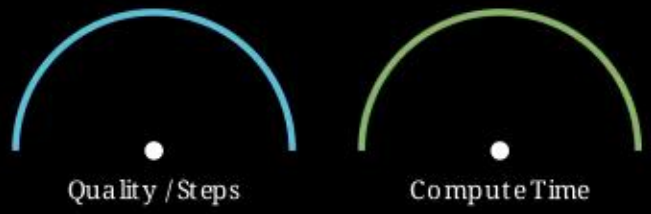
- Speed Optimization: Integrated Wan Tea Cache and distilled LoRA to bypass redundant inference.



# Mathematical Validation & Consistency

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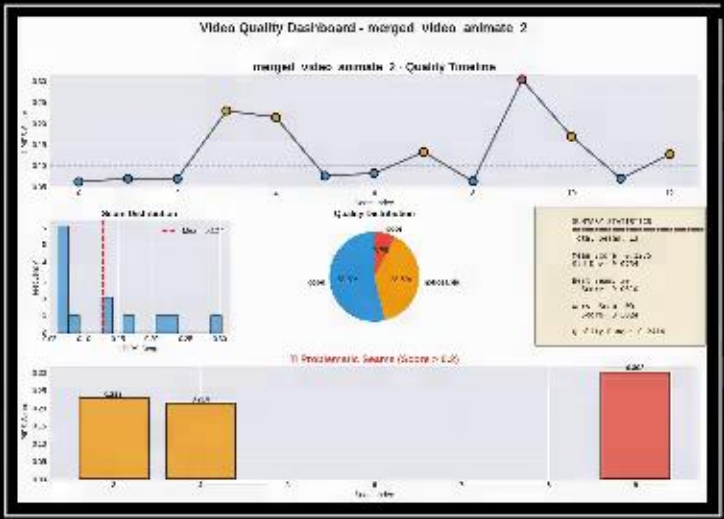

# Executive Summary & Roadmap

Current Status (Q2 2026):

✔ Finalized Long-Form (60s+) Workflow

✔ High-Fidelity 1080p Generation

✔ Perceptual Validation (LLIPS/LPIPS)

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## Future Directions:

Maintenance and Integrating it inside AI Agent Marketing Implementation

# Thank You!

Questions?

Full Documentation & PR: [https://github.com/Long-form-AI-video-generation/long-v2v-custom-node]